

Overview of Decision tree

Subject: Decision tree

1.

Overview A decision tree is a flowchart-like structure in which each internal node represents a test on an attribute (e.g. whether a coin flip comes up heads or tails), each branch represents the outcome of the test, and each leaf node represents a class label (decision taken after computing all attributes). The paths from root to leaf represent classification rules. In decision analysis, a decision tree and the closely related influence diagram are used as a visual and analytical decision support tool, where the expected values (or expected utility) of competing alternatives are calculated. A decision tree consists of three types of nodes:

2.

Once we have calculated the key metrics we can make some initial conclusions on the performance of the decision tree model built. The accuracy that we calculated was 71.60%. The accuracy value is good to start but we would like to get our models as accurate as possible while maintaining the overall performance. The sensitivity value of 19.64% means that out of everyone who was actually positive for cancer tested positive. If we look at the specificity value of 99.06% we know that out of all the samples that were negative for cancer actually tested negative. When it comes to sensitivity and specificity it is important to have a balance between the two values, so if we can decrease our specificity to increase the sensitivity that would prove to be beneficial. These are just a few examples on how to use these values and the meanings behind them to evaluate the decision tree model and improve upon the next iteration.

3.

Increasing the number of levels of the tree The accuracy of the decision tree can change based on the depth of the decision tree. In many cases, the tree's leaves are pure nodes. When a node is pure, it means that all the data in that node belongs to a single class. For example, if the classes in the data set are Cancer and Non-Cancer a leaf node would be considered pure when all the sample data in a leaf node is part of only one class, either cancer or non-cancer. A deeper tree is not always better when optimizing the decision tree. A deeper tree can influence the runtime in a negative way. If a certain classification algorithm is being used, then a deeper tree could mean the runtime of this classification algorithm is significantly slower. There is also the possibility that the actual algorithm building the decision tree will get significantly slower as the tree gets deeper. If the tree-building algorithm being used splits pure nodes, then a decrease in the overall accuracy of the tree classifier could be experienced. Occasionally, going deeper in the tree can cause an accuracy decrease in general, so it is very important to test modifying the depth of the decision tree and selecting the depth that produces the best results. To summarize, observe the points below, we will define the number D as the depth of the tree. Possible advantages of increasing the number D :

4.

Now, once we have chosen the root node we can split the samples into two groups based on whether a sample is positive or negative for the root node mutation. The groups will be called group A and group

B. For example, if we use M1 to split the samples in the root node we get NC2 and C2 samples in group A and the rest of the samples NC4, NC3, NC1, C1 in group B. Disregarding the mutation chosen for the root node, proceed to place the next best features that have the highest values for information gain or the phi function in the left or right child nodes of the decision tree. Once we choose the root node and the two child nodes for the tree of depth = 3 we can just add the leaves. The leaves will represent the final classification decision the model has produced based on the mutations a sample either has or does not have. The left tree is the decision tree we obtain from using information gain to split the nodes and the right tree is what we obtain from using the phi function to split the nodes.

5.

The decision tree illustrates that when sequentially distributing lifeguards, placing a first lifeguard on beach #1 would be optimal if there is only the budget for 1 lifeguard. But if there is a budget for two guards, then placing both on beach #2 would prevent more overall drownings.

6.

One major drawback of information gain is that the feature that is chosen as the next node in the tree tends to have more unique values. An advantage of information gain is that it tends to choose the most impactful features that are close to the root of the tree. It is a very good measure for deciding the relevance of some features. The phi function is also a good measure for deciding the relevance of some features based on "goodness". This is the information gain function formula. The formula states the information gain is a function of the entropy of a node of the decision tree minus the entropy of a candidate split at node t of a decision tree.

7.

if condition1 and condition2 and condition3 then outcome. Decision rules can be generated by constructing association rules with the target variable on the right. They can also denote temporal or causal relations.

8.

Optimizing a decision tree A few things should be considered when improving the accuracy of the decision tree classifier. The following are some possible optimizations to consider when looking to make sure the decision tree model produced makes the correct decision or classification. Note that these things are not the only things to consider but only some.

9.

Are simple to understand and interpret. People are able to understand decision tree models after a brief explanation. Have value even with little hard data. Important insights can be generated based on experts describing a situation (its alternatives, probabilities, and costs) and their preferences for outcomes. Help determine worst, best, and expected values for different scenarios. Use a white box model. If a given result is provided by a model. Can be combined with other decision techniques. The action of more than one decision-maker can be considered. Disadvantages of decision trees: